HEALTH360: COMPREHENSIVE HEALTHCARE RISK & DISEASE ANALYTICS

FINAL PROJECT REPORT
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Team ID: En25cs3010862
Project Name: Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks: Final Project Evaluation

1. INTRODUCTION

1.1 Project Overview

Health360 is an integrated, cloud-native healthcare analytics platform designed to transition modern medicine from reactive crisis treatment to proactive, preventive risk management. Leveraging Snowflake for scalable data warehousing, Python for data cleaning and ML risk modeling, and Tableau for interactive clinician dashboards, Health360 continuously ingests and analyzes patient vital signs, lifestyle metrics, and disease histories to provide early diagnostic risk indicators.

1.2 Purpose

The primary objective of Health360 is to empower medical practitioners, cardiologists, and health system administrators with timely, actionable insights. By predicting health risk escalation early, Health360 aims to reduce emergency readmission rates, optimize clinical resource allocation, and improve patient health outcomes.

2. IDEATION PHASE

2.1 Problem Statement

Traditional healthcare systems heavily rely on periodic, reactive clinical visits. Vital signs and physiological risk markers are often recorded as static snapshots in disparate paper or legacy systems. Consequently, gradual degradation in patient health or sudden vital anomalies pass unnoticed until acute medical emergencies occur.

2.2 Empathy Map Canvas

The Empathy Map captures the dual perspectives of primary platform stakeholders (Lead Cardiologists & Chronic Care Patients):

SAYS • "I need a quick way to review patient vitals without digging through paper records." • "Alerts need to be accurate so we don't experience alarm fatigue."	THINKS • "Are my patients complying with medication and exercise regimens?" • "I worry about sudden cardiac or diabetic emergencies happening unobserved."
PERSONA NAME Dr. Rajesh Sharma (Lead Cardiologist)	PERSONA NAME Ramesh Kumar (Chronic Patient, Age 58)
DOES • Reviews daily vital sign trends on web dashboards. • Manages patient triage based on automated AI disease risk scores.	FEELS • Anxious about unpredictable health flare-ups. • Reassured when continuous monitoring tools provide timely feedback.

2.3 Brainstorming

During ideation, multiple architectural approaches were evaluated. The team prioritized an end-to-end cloud pipeline combining IoT device telemetry, Snowflake automated data warehousing, Python ETL scripts for median imputation and bounded validation, and Tableau visual storytelling.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage 1 (Onboarding & Setup): Patient registers profile and pairs IoT wearables via Bluetooth.
Stage 2 (Continuous Monitoring & Triage): IoT sensors stream vitals to Snowflake; automated engines compute composite risk scores and trigger alerts.
Stage 3 (Intervention & Wellness): Clinicians review risk dashboards, conduct video/in-person consults, and adjust personalized wellness plans.

3.2 Solution Requirement

• Functional Requirements: Ingestion of 5,000 patient records, null imputation, composite risk scoring (0-100 scale), role-based Tableau filtering, and alert generation.
• Non-Functional Requirements: Query latency < 0.5s, 99.9% uptime, role-based access control (RBAC), and HIPAA-compliant data security.

3.3 Data Flow Diagram

Level 0 DFD: Patient IoT Vitals & Clinical Data -> Health360 System -> Clinician Dashboards & Alert Engine.
Level 1 DFD: Ingestion (RAW) -> Data Transformation & Imputation (STAGING) -> Risk Modeling & Analytics (CURATED) -> Visual Reporting (MART).

3.4 Technology Stack

• Database & Data Warehouse: Snowflake DB (RAW, STAGING, CURATED, MART schemas)
• Data Engineering & ML: Python 3.10, Pandas, NumPy
• Business Intelligence & Visualization: Tableau Desktop / Tableau Cloud
• Cloud & Storage: Snowflake Virtual Warehouse (MEDIUM size)

4. PROJECT DESIGN

4.1 Problem Solution Fit

Health360 addresses the core root cause—reactive healthcare delivery—by delivering continuous, predictive risk scoring that fits seamlessly into clinical workflows.

4.2 Proposed Solution

A multi-tiered healthcare monitoring platform that processes vital signs (Heart Rate, SpO2, Blood Pressure, Glucose) and lifestyle factors (Exercise Level, Smoking Status) to auto-classify patients into Low, Moderate, and High Risk tiers.

4.3 Solution Architecture

5-Layer Architecture: 1. Ingestion Layer (IoT Sensors & CSV) -> 2. Staging Layer (Snowflake ETL) -> 3. Analytics Layer (Python Risk Engine) -> 4. Visualization Layer (Tableau) -> 5. Decision Support Layer (Clinician Triage).

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Executed across 4 Sprints: Sprint 1 (Data Ingestion & Snowflake Setup), Sprint 2 (ETL Transformation & Data Cleansing), Sprint 3 (Risk Scoring Model & Tableau Dashboard Development), Sprint 4 (Performance Testing & Final Documentation).

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

• Snowflake Performance: Query execution times ranged between 0.12s and 0.38s on a 5,000-record dataset (~1.2 MB).
• Tableau Performance: Dashboard rendering time < 0.5s with real-time interactive global filtering across 8 distinct visualizations.

7. RESULTS

7.1 Output Screenshots & Summary

• Total Patient Cohort Analyzed: 5,000 Records
• Diabetes Rate: 34.9%
• Heart Disease Rate: 30.0%
• Stroke Rate: 24.9%
• Average BMI: 24.97 | Average Blood Pressure: 134.4 mmHg
• Insights: Highest chronic disease prevalence concentrated in the 65+ age demographic; regular exercise level 2 lowers mean risk scores from 49.2 to 34.1.

8. ADVANTAGES & DISADVANTAGES

Advantages:
• Enables proactive clinical intervention before disease escalation.
• High-performance, scalable Snowflake cloud infrastructure.
• Intuitive Tableau dashboards providing clear visual risk segmentation.

Disadvantages:
• Requires continuous network connectivity for IoT live streaming.
• Initial hardware costs for high-precision diagnostic wearables.

9. CONCLUSION

Health360 successfully demonstrates the power of combining modern cloud data warehousing (Snowflake), statistical data processing (Python), and business intelligence (Tableau) to deliver a robust, highly responsive healthcare risk analytics platform.

10. FUTURE SCOPE

• Phase 2: Live IoT telemetry stream integration via MQTT and Apache Kafka.
• Phase 3: Advanced Deep Learning models (LSTM/Transformer) for early sepsis and cardiac arrest forecasting.
• Phase 4: Full HL7/FHIR EMR interoperability and mobile push notification alerts.

11. APPENDIX

Dataset Link: Kaggle Smart Healthcare Dataset (~5,000 anonymized patient records)

GitHub Link: https://github.com/shwetasinghchauhan1110/Health360-Comprehensive-Healthcare-Risk-Disease-Analytics.git